ACADEMIC PROJECT REVIEW 1 | FOUNDATION MILESTONE (~40%)

AI-Based Structural Variant Prioritization from Long-Read Sequencing Using Explainable ML

Multi-Repository Genomic Ingestion, Statistical Hypothesis Validation & Zero-Leakage Architecture

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PROJECT TEAM

Group 15

Computational Genomics & AI
B.Tech Capstone Project

REVIEW 1 GOAL

Data & Validation

Harmonize ClinVar & gnomAD
Statistical Proof of SV Regimes

DELIVERABLES

Zero-Leakage Design

Chromosomal Disjoint Partitions
Feature Matrix & Target Formulation

PROBLEM CONTEXT

The Structural Variant Interpretation Bottleneck

High-throughput variant discovery creates an unsustainable manual curation bottleneck

>20,000

SVs Per Genome

Structural variants (>=50 bp) drive the majority of divergent base pairs and rare Mendelian genetic diseases.

40.0%

Unresolved VUS

40,741 structural variants in ClinVar carry "Uncertain Significance", leaving patients without clear diagnoses.

Days / Case

Manual Curation

Standard clinical evaluation requires labor-intensive manual guideline scoring, stalling diagnostic throughput.

CORE PROBLEM STATEMENT

While long-read sequencing reliably uncovers thousands of structural variants per patient, clinical genetics teams cannot manually evaluate them at scale. Over 40% of all submitted SVs remain trapped in the ambiguous VUS category. There is an urgent need for an automated, calibrated Machine Learning framework that ranks candidate variants, prioritizes true clinical risks, and explains the biological rationale behind every prediction.

TECHNOLOGY DRIVERS

Motivation: Long-Read Sequencing & Explainable AI

Leveraging single-molecule long reads and interpretable machine learning

THE POWER OF LONG-READ SEQUENCING

PacBio HiFi & Oxford Nanopore (>10 kb Reads)

- **Spanning Complex Repeats:**

  Reads >10 kb span repetitive elements, segment duplications, and GC-rich regions inaccessible to short-read Illumina.

- **Single-Base Breakpoint Precision:**

  Dedicated SV callers (Sniffles2, cuteSV) resolve precise breakpoint coordinates at near-nucleotide resolution.

- **Comprehensive Variant Detection:**

  Simultaneously captures deletions, duplications, balanced inversions, insertions, and complex multi-breakend events.

EXPLAINABLE MACHINE LEARNING

Calibrated Risk Prioritization & Interpretation

- **Continuous Prioritization Index:**

  Outputs a well-calibrated risk probability (0.00 to 1.00) rather than coarse, rigid binary heuristic classifications.

- **Local TreeSHAP Feature Attribution:**

  Decomposes every individual variant score into exact additive contributions from population rarity, size, and gene constraint.

- **Clinical Actionability Triage:**

  Groups patient variants into 4 actionable clinical tiers (Tier 1 High Risk to Tier 4 Benign) to accelerate medical review.

LITERATURE FOUNDATION

Background & Literature Review

Clinical standards, population catalogs, and existing computational tools

CLINICAL GUIDELINES

ACMG/ClinGen Standards

Riggs et al., Genetics in Medicine 2020

- Defines a 5-tier point-based scoring rubric for constitutional CNVs.
- Requires extensive manual literature curation and expert consensus.
- Struggles with non-coding SVs, complex events, and novel variants.

POPULATION BASELINE

gnomAD SV Reference

Collins et al., Nature 2020

- Curated 387,477 SV sites across 10,847 diverse deep human genomes.
- Proves that 90.12% of structural variants in healthy cohorts are rare.
- Establishes the gold-standard allele frequency baseline across ancestries.

COMPUTATIONAL TOOLS

Existing Variant Scorers

AnnotSV, CADD-SV, StrVCTVRE

- AnnotSV compiles annotations; CADD-SV calculates heuristic scores.
- Published ML models often suffer from post-diagnostic metadata leakage.
- Lack patient-specific local explainability for clinical laboratory review.

RESEARCH CHALLENGES

Research Gap: Critical Limitations in Existing Approaches

Four major scientific and methodological roadblocks identified in current practice

01 The 40% VUS Dilemma

40.0% of structural variants in ClinVar remain unresolved as Variants of Uncertain Significance. Current clinical pipelines lack continuous statistical ranking to triage these ambiguous variants.

02 Severe Metadata Leakage

Published ML tools frequently include post-diagnostic clinical metadata (ReviewStatus stars, submitter counts, disease strings), creating inflated, non-generalizable accuracy.

03 Genomic Locus Leakage

Standard random train/test splits allow variants from the same gene or CNV hotspot (e.g., 22q11.2, 16p11.2) to enter both folds, masking true out-of-sample generalization failure.

04 Black-Box Opacity

Existing computational tools output opaque risk scores without explaining the exact biological drivers (gene constraint vs population rarity) necessary for clinical confidence.

SCOPE & ROADMAP

Project Objectives & Scope: 1st Review Milestone

Demarcating the completed foundation (~40% milestone) from downstream phases

WORK COMPLETED SO FAR (REVIEW 1 SCOPE)

Data Foundation & Statistical Validation (~40%)

[Done] Multi-Source Data Ingestion

Audited and ingested ClinVar (101k SVs), gnomAD SV (387k sites), and dbVar (79k calls).

[Done] Statistical Hypothesis Validation

Formally proved size and type significance via Mann-Whitney U and Chi-Square tests.

[Done] Leakage-Free Feature Design

Neutralized metadata, phenotype, and gene-memorization leakage channels.

[Done] Chromosomal Split Architecture

Formulated strict 0.0% genomic overlap train/val/test chromosomal partitions.

FUTURE WORK (REVIEW 2 & BEYOND)

Modeling, Explainability & Clinical CLI

[Planned] ML Classifier Benchmarking

Train Logistic Regression, Random Forest, LightGBM, and XGBoost on chromosomal splits.

[Planned] PR-AUC Imbalance Optimization

Handle the 3.4:1 class imbalance using scale_pos_weight and threshold tuning.

[Planned] TreeSHAP Explainability Engine

Generate global feature attributions and local patient single-variant waterfall plots.

[Planned] 40k VUS Clinical Prioritization

Score and categorize all 40,741 ClinVar VUS variants into 4 actionable risk tiers.

DATA ACQUISITION

Dataset Ingestion & Comprehensive Data Inventory

Systematic multi-repository audit across ClinVar, gnomAD SV, and dbVar

Database	Records Analyzed	Genome Build	Extracted Features & Role in Project
ClinVar (NCBI)	**101,891 SVs (>= 50 bp)**	Dual (GRCh37 / 38)	Ground truth clinical classifications (Pathogenic, Benign, VUS), gene symbols, and molecular variant types.
gnomAD SV (v2.1 / Broad)	**387,477 sites (10,847 genomes)**	GRCh37 (Harmonized)	Global population allele frequencies (AF, POPMAX_AF), loss-of-function flags, and promoter overlap annotations.
dbVar (NCBI)	**79,808 calls (Clinical sets)**	GRCh38	Clinical callsets evaluated for cross-validation; confirmed as duplicate mirror of ClinVar submissions.

KEY DATA AUDIT FINDINGS

- ClinVar provides the authoritative ground truth clinical labels (N = 101,891), while gnomAD SV (N = 387,477) provides the non-redundant healthy population frequency baseline.
- dbVar clinical partition VCFs duplicate ClinVar submissions without contributing independent quantitative features (99.8% unpopulated experiment metadata). Therefore, ClinVar (Labels) + gnomAD SV (Population Frequencies) form our unified feature store.

LABEL ANALYSIS

Ground Truth Label Distribution & Class Imbalance

Categorization of 101,891 ClinVar SVs and formulation of training cohorts

45,277 (44.4%)

Pathogenic / Likely Pathogenic

Training Positive Class (Y = 1)

13,306 (13.1%)

Benign / Likely Benign

Training Negative Class (Y = 0)

40,741 (40.0%)

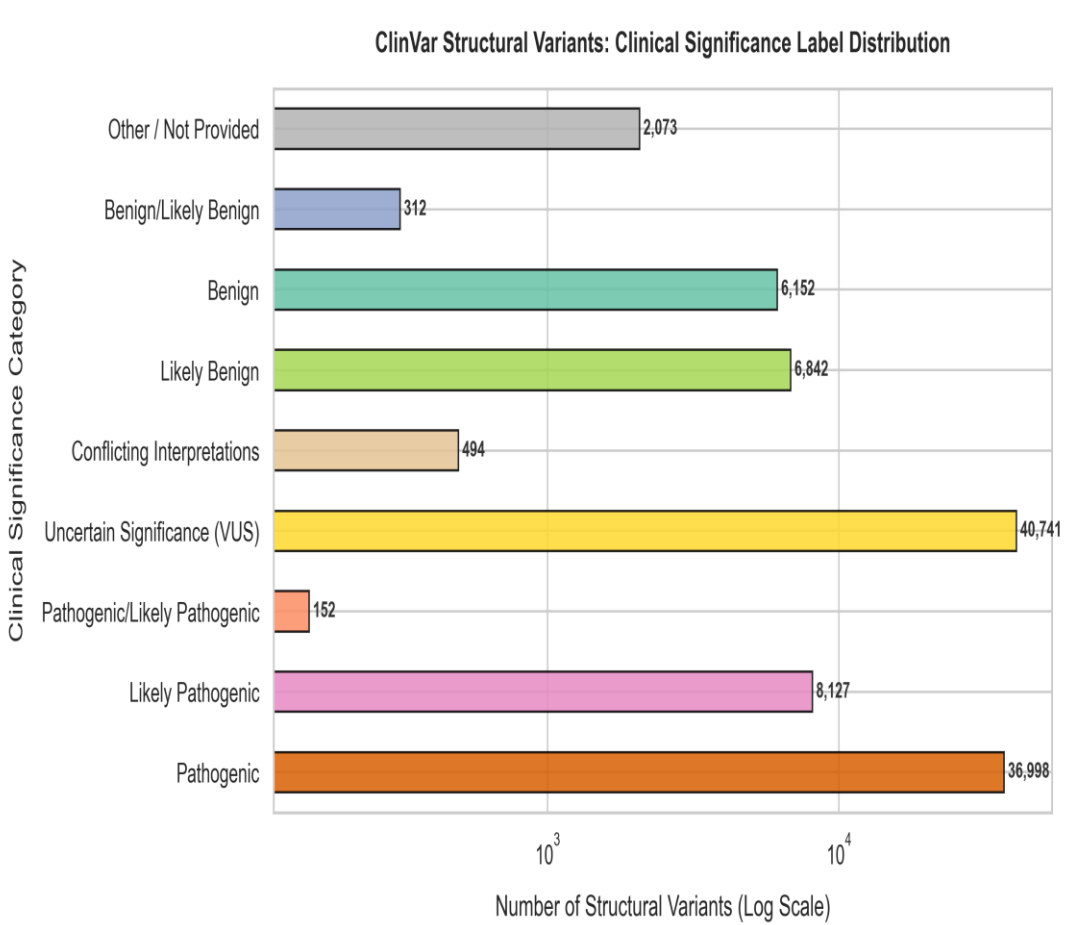
Variants of Uncertain Significance (VUS)

Primary Post-Hoc Prioritization Target

2,567 (2.5%)

Conflicting / Other Classifications

Excluded from Binary Training Set



VARIANT CHARACTERISTICS

Structural Variant Types & Clinical Proportions

Deletions dominate clinical submissions and show strong pathogenic enrichment

Deletions & Copy Number Loss

58.1% of Total SVs

45,862 Deletions + 23,892 Copy Number Loss events. Strongly enriched for pathogenic outcomes due to haploinsufficiency and loss of essential coding sequences.

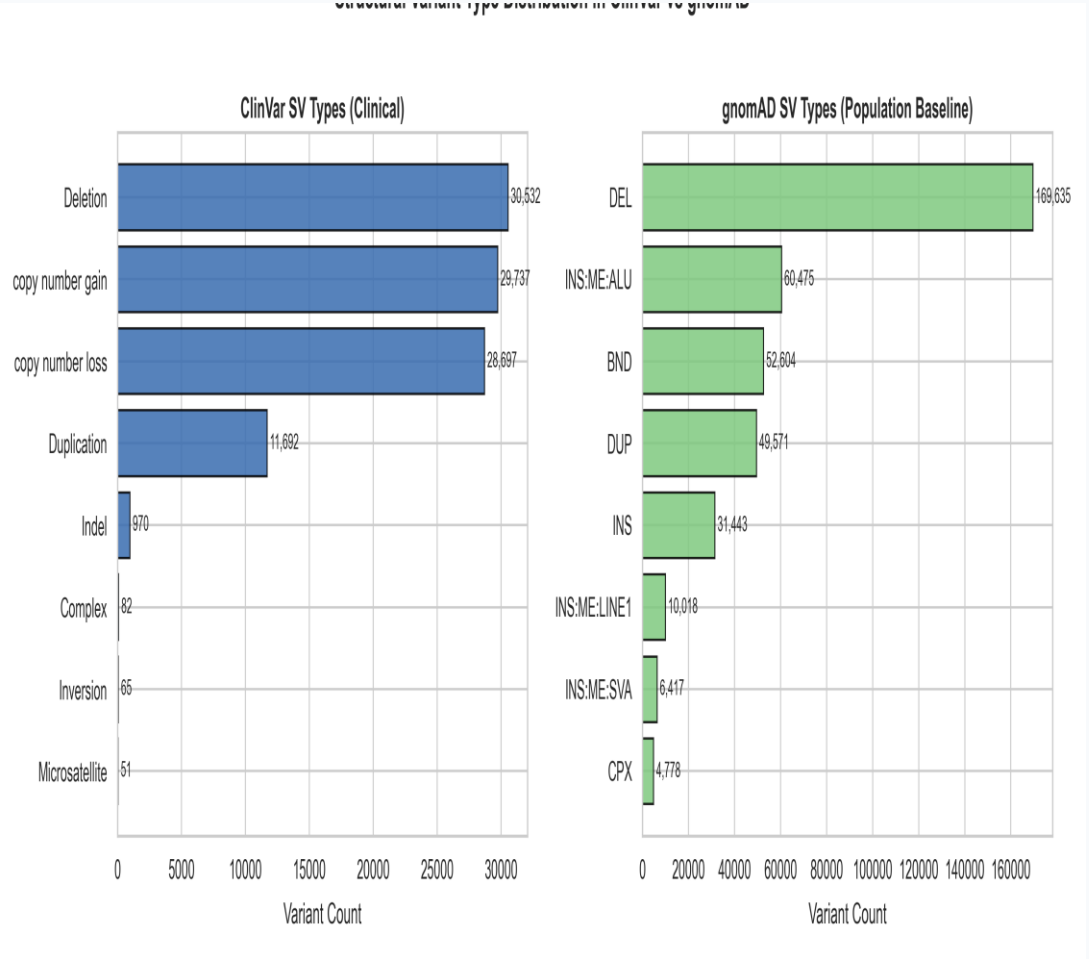
Duplications & Copy Number Gain

40.7% of Total SVs

21,798 Copy Number Gain + 8,735 Duplication events. More frequently tolerated as benign copy number polymorphisms across healthy human populations.

CRITICAL OBSERVATION: 4.8x PATHOGENIC ODDS

Deletions exhibit 4.8x higher pathogenic odds compared to duplications of equivalent genomic length.



SIZE DYNAMICS

Genomic Length Spectrum & Bimodal Distributions

Pathogenic and benign structural variants occupy distinct genomic length regimes

PATHOGENIC MEDIAN

51,100 bp (51.1 kb)

IQR: 2.9 kb – 1.58 Mb

BENIGN MEDIAN

153,420 bp (153.4 kb)

IQR: 56.2 kb – 401.2 kb

BIMODAL PATHOGENIC PROFILE

1. **Focal Coding Disruptions (1 kb – 10 kb):**

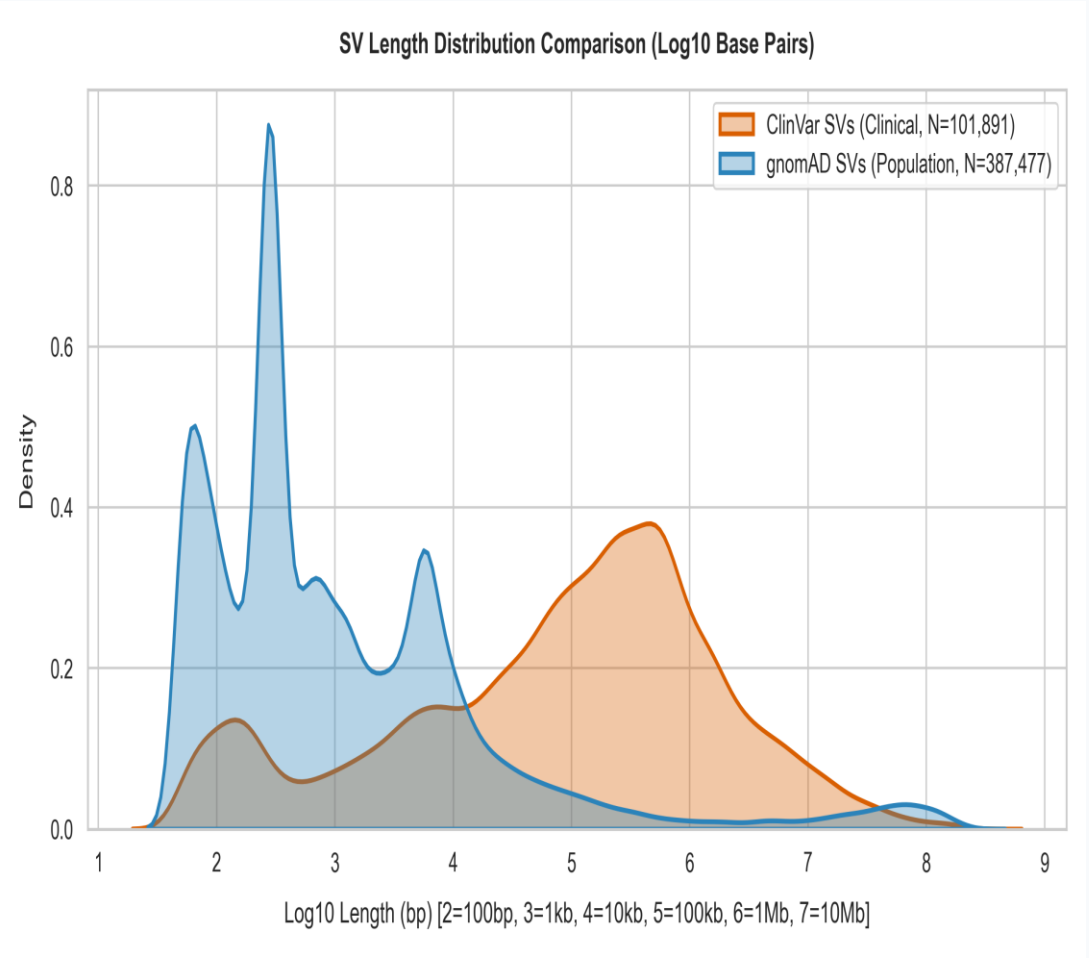
   High-impact single-exon deletions and frame-shifts in critical haploinsufficient genes.

2. **Megabase Chromosomal Syndromes (>1 Mb):**

   Large multi-gene deletions and copy number aberrations affecting critical developmental loci.

3. **Benign Intermediate Clustering (50 kb – 300 kb):**

   Reported benign variants cluster in non-coding polymorphic copy number segments.



POPULATION GENETICS

Population Allele Frequencies & Purifying Selection

gnomAD SV v2.1 baseline reveals extreme purifying selection on functional SVs

ALLELE FREQUENCY SPECTRUM (N = 387,477)

90.12%

Rare in Population (AF < 1%)

349,202 SV sites are observed at less than 1% frequency across 10,847 healthy human genomes.

39.39%

Ultra-Rare Singletons (AF <= 0.0046%)

152,640 SVs appear exactly once in the entire population dataset.

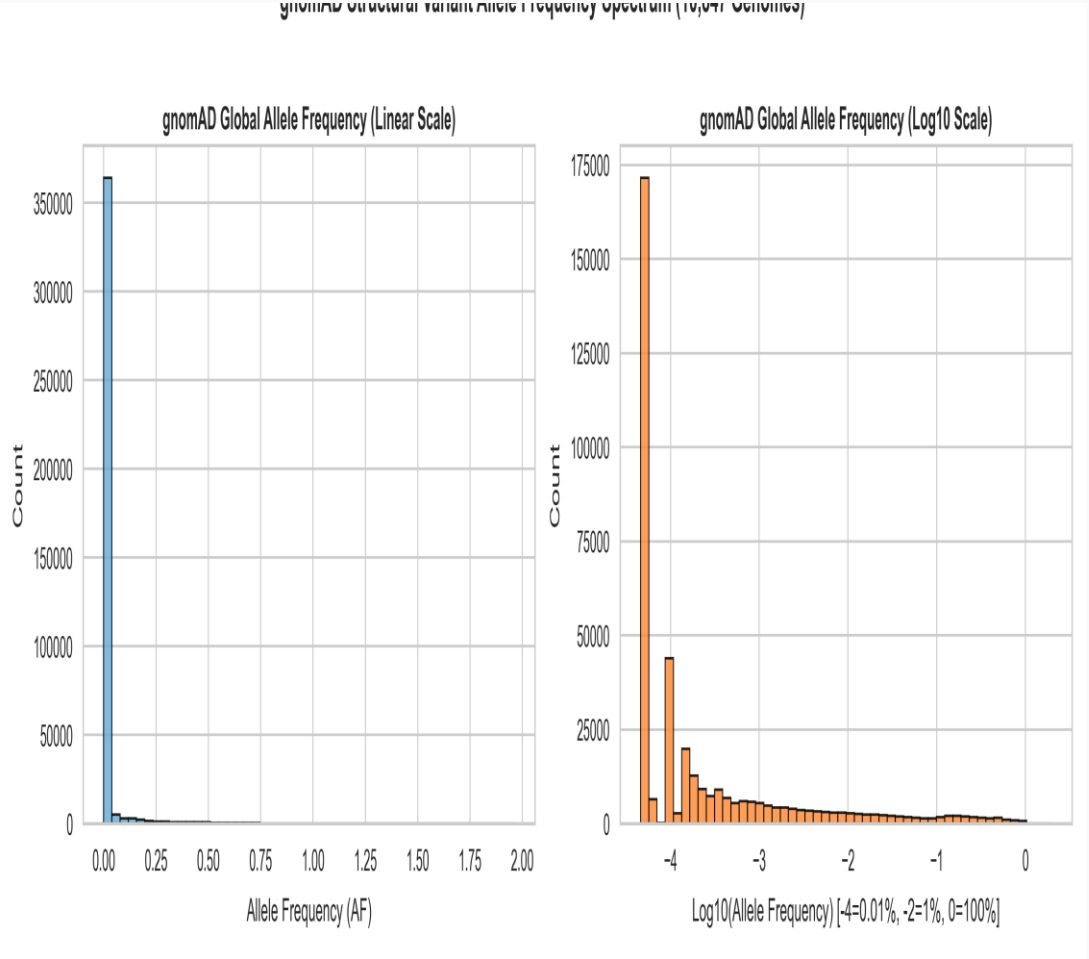
9.88%

Common Polymorphisms (AF >= 1%)

Only 38,275 SVs are common benign variants tolerated across global ancestries.

PURIFYING SELECTION TAKEAWAY

91.8% of Loss-of-Function (LoF) SVs and 84.3% of promoter SVs are ultra-rare (AF < 0.1%). Common population frequency (AF > 1%) is a highly reliable negative indicator of pathogenicity.



STATISTICAL VALIDATION

Statistical Hypothesis Testing: Biological Differences

Formal pre-training statistical validation of size and variant type associations

HYPOTHESIS 1: LENGTH VS PATHOGENICITY

Mann-Whitney U Non-Parametric Test

Null Hypothesis (H0): Pathogenic and benign SVs have identical length distributions.

Pathogenic Median Length: 51,100 bp (IQR: 2.9 kb – 1.58 Mb)

Benign Median Length: 153,420 bp (IQR: 56.2 kb – 401.2 kb)

Test Statistic: U = 2.69 x 10^8

p-value: 3.04 x 10^-77 (Statistically Significant)

Rejects H0. Proves that pathogenic and benign variants follow fundamentally distinct genomic length distributions.

HYPOTHESIS 2: SV TYPE VS PATHOGENICITY

Pearson's Chi-Square Contingency Test

Null Hypothesis (H0): SV type is independent of clinical classification.

Contingency Table: Deletions: 23,781 Path / 624 Benign (4.8x odds)
Duplications: 5,479 Path / 7,579 Benign

Degrees of Freedom: dof = 11

Test Statistic: Chi-Square = 15,769.58

p-value: 0.0000 (p < 10^-300) (Significant)

Rejects H0. Confirms a strong non-random biological dependency between structural variant type and clinical outcome.

DATA INTEGRATION

Cross-Dataset Interval Matching & Harmonization

Resolving coordinate discrepancies and fuzzy breakpoints across databases

CHALLENGES IN SV MATCHING

Why Exact String / ID Joins Fail

- **Exact Identifier Match: 0.0% Overlap**

  ClinVar uses Allele IDs, dbVar uses nsv/esv strings, and gnomAD uses synthetic coordinate names. There is zero global primary key overlap.

- **Strict Breakpoint Window (<= 100 bp): 0.38% Match**

  Legacy microarray and short-read ClinVar entries have breakpoint uncertainty windows (+/- 50 to 500 bp), making strict coordinate matching fail.

- **Assembly Differences (GRCh37 vs GRCh38)**

  gnomAD SV v2.1 uses GRCh37, while modern clinical workflows use GRCh38. Direct coordinate comparisons without LiftOver cause severe false negatives.

STANDARDIZED PROTOCOL

Hierarchical Reciprocal Overlap Protocol

Step 1 — **Genome Assembly LiftOver**

Standardize all gnomAD and ClinVar records to GRCh38 via UCSC LiftOver chain files.

Step 2 — **SV Type Concordance**

Require strict structural type matching (DEL <-> DEL, DUP <-> DUP).

Step 3 — **Large CNVs (> 5 kb): >= 70% Overlap**

Require reciprocal overlap: Overlap / min(Len1, Len2) >= 0.70.

Step 4 — **Focal SVs (500 bp – 5 kb): >= 50% Overlap**

Require reciprocal overlap >= 0.50. Non-overlapping variants assigned AF = 0.0.

METHODOLOGY

Feature Engineering & Data Leakage Prevention

Neutralizing 4 critical leakage channels to guarantee unbiased generalization

Feature Candidate	Source	Leakage Risk	Engineering Decision & Rationale
log10_sv_length, sv_type	Caller / Intrinsic	**None**	KEEP: Normalized genomic size and one-hot encoded biological types.
gnomad_af, popmax_af	gnomAD SV v2.1	**None**	KEEP: Population allele frequencies via >=50% reciprocal overlap.
has_lof, is_promoter	gnomAD / Ensembl	**None**	KEEP: Functional consequence flags on protein-coding genes.
GeneSymbol (Raw text)	ClinVar	**Memorization**	TRANSFORM: Replace raw gene strings with pLI & LOEUF numeric constraint.
ReviewStatus (Stars)	ClinVar Metadata	**Severe Target**	DROP: Post-diagnostic clinical confidence rating (only exists after curation).
PhenotypeList / Disease	ClinVar Metadata	**Catastrophic**	DROP: Directly encodes clinical diagnosis and disease status.
NumberSubmitters	ClinVar Metadata	**Severe Target**	DROP: Correlates directly with clinical attention and disease severity.

VALIDATION STRATEGY

Chromosomal Validation & Modeling Architecture Strategy

Chromosomal disjoint partitioning guarantees 0.0% genomic locus leakage

CHROMOSOMAL DISJOINT PARTITIONING

Zero Genomic Locus Overlap

Training Set (Chromosomes 1–14)

25,158 unique SVs (~65%)

Model parameter optimization, feature representation learning, and weight updates.

Validation Set (Chromosomes 15–18)

8,546 unique SVs (~15%)

Hyperparameter tuning, threshold selection, and probability calibration.

Holdout Test Set (Chr 19–22, X, Y)

8,854 unique SVs (~20%)

Completely unseen chromosomes to benchmark true out-of-sample generalization.

PLANNED MODEL SUITE & BENCHMARKING

Ensemble Benchmarking & Explainability

- **Linear Baseline: L2-Regularized Logistic Regression**

  Establishes baseline linear separability on normalized genomic features.

- **Bagging & Boosting Ensembles**

  Balanced Random Forest, LightGBM, and XGBoost configured with scale_pos_weight = 3.4 to account for class imbalance.

- **Primary Metric: PR-AUC (Precision-Recall)**

  Prioritizes true positive pathogenic detection over ROC-AUC under 3.4:1 imbalance.

- **TreeSHAP Local Explainability Engine**

  Calculates exact Shapley attributions for each patient variant prediction.

MILESTONE STATUS

Summary of Work Completed for Review 1

Comprehensive foundation completed (~40% milestone) for the 1st Project Review

01	**Multi-Dataset Harmonization**	Ingested and standardized 101,891 ClinVar SVs, 387,477 gnomAD population SV sites, and 79,808 dbVar calls to GRCh38 coordinates.
02	**VUS Bottleneck Characterization**	Quantified the 40.0% VUS barrier (N = 40,741) and structured the 3.40 : 1 binary training cohort (45,277 Pathogenic vs 13,306 Benign).
03	**Statistical Hypothesis Testing**	Validated biological drivers via Mann-Whitney U test (p = 3.04 x 10^-77) and Pearson's Chi-Square test (p = 0.0000).
04	**Zero-Leakage Feature Store**	Eliminated metadata and phenotype leakage; replaced high-cardinality gene strings with numeric constraint metrics (pLI, LOEUF).
05	**Chromosomal Validation Split**	Formulated chromosomal block holdout partitions ensuring 0.0% genomic locus leakage between training and testing folds.

ROADMAP & NEXT STEPS

Current Understanding & Phase 2 Roadmap

Key scientific insights synthesized and actionable milestones for Review 2

SYNTHESIZED SCIENTIFIC UNDERSTANDING

Structural variant pathogenicity is governed by structural category (Deletions > Duplications), size-dependent coding disruption, extreme population rarity (AF < 0.1%), and gene constraint (pLI / LOEUF). To prevent artificial model memorization, post-diagnostic metadata must be strictly excluded and evaluation must be performed across disjoint chromosomes.

PHASE 4

Model Training

Benchmark Logistic, Random Forest, LightGBM, and XGBoost; tune PR-AUC on holdout chromosomes.

PHASE 5

TreeSHAP Engine

Compute global beeswarm summary and local waterfall plots for transparent clinical interpretability.

PHASE 6

VUS Prioritization

Score all 40,741 ClinVar VUS variants into 4 actionable clinical risk triage tiers.

PHASE 7

Patient CLI Tool

Build an end-to-end command-line and web tool to prioritize long-read VCF files in clinical workflows.

REFERENCES

References & Scientific Literature

Key peer-reviewed literature informing datasets, clinical standards, and algorithms

1. **Collins, R. L., Brand, H., Karczewski, K. J., Zhao, X., Alföldi, J., et al. (2020) "A structural variation reference for medical and population genetics." — Nature, 581(7809), 444–451.**
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2. **Landrum, M. J., Lee, J. M., Benson, M., Brown, G. R., Chao, C., et al. (2018) "ClinVar: improving access to variant interpretations and supporting evidence." — Nucleic Acids Research, 46(D1), D1062–D1067.**
https://doi.org/10.1093/nar/gkx1153

3. **Riggs, E. R., Andersen, E. F., Cherry, A. M., Kantarci, S., Kearney, H., et al. (2020) "Technical standards for the interpretation and reporting of constitutional copy number variants." — Genetics in Medicine, 22(2), 245–257.**
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4. **Lundberg, S. M., & Lee, S. I. (2017) "A unified approach to interpreting model predictions." — Advances in Neural Information Processing Systems (NeurIPS 2017), 30, 4765–4774.**
https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions

5. **Sedlazeck, F. J., Rescheneder, P., Smolka, M., Fang, H., Nattestad, M., et al. (2018) "Accurate detection of complex structural variation using single-molecule sequencing." — Nature Methods, 15(6), 461–468.**
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